

Open Science in Musculoskeletal Imaging: From Principles to Practice

Serena Bonaretti

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July 8, 2026

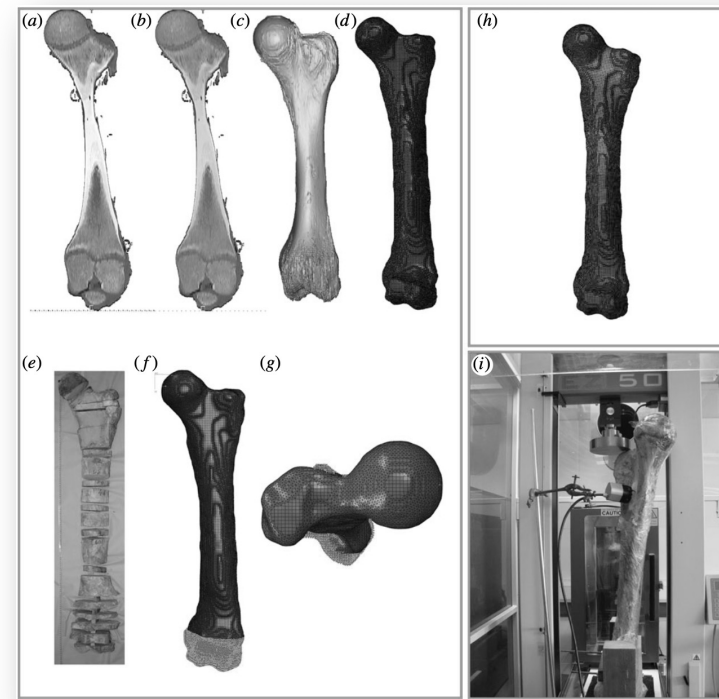
Why open science?

What makes an accurate and reliable subject-specific finite element model? A case study of an elephant femur

O. Panagiotopoulou , S. D. Wilshin, E. J. Rayfield, S. J. Shefelbine and J. R. Hutchinson

Published: 13 July 2011 | <https://doi.org/10.1098/rsif.2011.0323>

In this study, we seek to construct the first reliable FE model of the femur of an adult Asian elephant (*Elephas maximus* Linnaeus 1758) in order to move towards a better understanding of its form–function relationship and re-examine previous ideas about bone strength, scaling and athleticism in these giant land animals [10,11]. We first test our model's precision by conducting a series of convergence tests with regards to the size and type of elements used in the geometric mesh. We then perform a sensitivity study on the model by determining the effects of varying its material properties. Finally, we compare our FEA results against the experimental data obtained under prescribed loading conditions in the laboratory (using laser speckle interferometry). We hope that our study will raise the methodological standards in zoological applications of FEA by encouraging the inclusion of convergence tests, sensitivity studies and validation analyses to ensure model reliability.



What makes an accurate and reliable subject-specific finite element model? A case study of an elephant femur

O Panagiotopoulou, [SD Wilshin](#), [EJ Rayfield](#), [SJ Shefelbine](#), JR Hutchinson

Journal of the Royal Society Interface, 2012 • [royalsocietypublishing.org](#)

Finite element modelling is well entrenched in comparative vertebrate biomechanics as a tool to assess the mechanical design of skeletal structures and to better comprehend the complex interaction of their form–function relationships. But what makes a reliable subject-specific finite element model? To approach this question, we here present a set of convergence and sensitivity analyses and a validation study as an example, for finite element analysis (FEA) in general, of ways to ensure a reliable model. We detail how choices of element size, type and material properties in FEA influence the results of simulations. We also present an empirical model for estimating heterogeneous material properties throughout an elephant femur (but of broad applicability to FEA). We then use an *ex vivo* experimental validation test of a cadaveric femur to check our FEA results and find that the heterogeneous model matches the experimental results extremely well, and far better than the homogeneous model. We emphasize how considering heterogeneous material properties in FEA may be critical, so this should become standard practice in comparative FEA studies along with convergence analyses, consideration of element size, type and experimental validation. These steps may be required to obtain accurate models and derive reliable conclusions from them.

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- In some papers, the citation is in the literature review
- In other papers, the findings are used to build upon

2011

2026

What makes an accurate and reliable subject-specific finite element model? A case study of an elephant femur

O Panagiotopoulou, SD Wilshin, EJ Rayfield, SJ Shefelbine and JR Hutchinson

Journal of the Royal Society Interface, 2012

[https://doi.org/10.1098/rsif.2012.0055](#)

In this study, we seek to construct the first reliable FEA model of the femur of an adult Asian elephant (*Elephas maximus* Linnaeus 1758) in order to assess

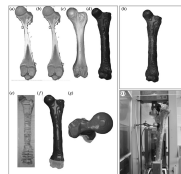
material properties and mechanical performance of the bone structure

throughout the bone. We use a series of convergence and sensitivity analyses to assess the influence of element size, type and material properties on the results of the FEA.

We then use an *ex vivo* experimental validation test of a cadaveric femur to check our FEA results and find that the heterogeneous model matches the experimental results extremely well, and far better than the homogeneous model.

We emphasize how considering heterogeneous material properties in FEA may be critical, so this should become standard practice in comparative FEA studies along with convergence analyses, consideration of element size, type and experimental validation.

These steps may be required to obtain accurate models and derive reliable conclusions from them.



Testing hypotheses for the function of the carnivoran baculum using finite-element analysis

Charlotte A. Brassey¹, James D. Gardiner² and Andrew C. Kitchener³

Accepted: 28 August 2018

mesh' options. Four-node linear tetrahedra were preferred for FEA as previous research has found such meshes perform well when compared with *ex vivo* experimental strain magnitudes [24]. Furthermore, FEA strain values have been found to converge when element numbers exceed 200 000 [24] to 800 000 [19] elements. Here, meshes comprised on average 2 million elements (mean 2 150 467, range 288 326–5 527 826). Meshes were exported from AVIZO in the proprietary ABAQUS FEA solver (Dassault Systems, Paris) file format.

24. Panagiotopoulou O, Wilshin SD, Rayfield EJ, Shefelbine SJ, Hutchinson JR. 2012 What makes an accurate and reliable subject-specific finite element model? A case study of an elephant femur. *J. R. Soc. Interface* **9**, 351–361. (doi:10.1098/rsif.2011.0323)

2011

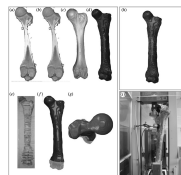
2018

2026

What makes an accurate and reliable subject-specific finite element model? A case study of an elephant femur

© Panagiotopoulou O, Wilshin SD, Rayfield EJ, Shefelbine SJ, Hutchinson JR. 2012. Published by the Royal Society.

In this study, we seek to construct the first reliable FE model of the femur of an adult African elephant (*Elephas maximus* Linnaeus 1758) in order to assess the mechanical function of the bone. We compare the results of two different meshing strategies and validation techniques to those of a single meshing strategy and validation technique. We then compare the results of two different meshing strategies and validation techniques to those of a single meshing strategy and validation technique. We then compare the results of two different meshing strategies and validation techniques to those of a single meshing strategy and validation technique.



What makes an accurate and reliable subject-specific finite element model? A case study of an elephant femur

© Panagiotopoulou O, Wilshin SD, Rayfield EJ, Shefelbine SJ, Hutchinson JR. 2012. Published by the Royal Society.

Finite element modelling is well established in comparative vertebrate biomechanics as a tool to assess the mechanical function of animal structures and to better understand the complex interaction of their form-function relationships. But what makes a reliable subject-specific finite element model? To address this question, we have assessed a set of 100 subject-specific finite element models (FEA) in general, of which we focus on a single model: the adult femur of an African elephant (*Elephas maximus* Linnaeus 1758). We compare the results of two different meshing strategies and validation techniques to those of a single meshing strategy and validation technique. We then compare the results of two different meshing strategies and validation techniques to those of a single meshing strategy and validation technique.

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What makes an accurate and reliable subject-specific finite element model? A case study of an elephant femur

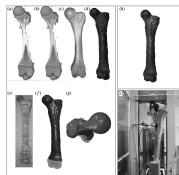
Published: 06 October 2014 | <https://doi.org/10.1098/rsif.2014.0854>

J. R. Soc. Interface **9**, 351–361 (7 February 2012; Published online 13 July 2011)
(doi:10.1098/rsif.2011.0323)

In the analysis presented in this paper, finite element model nodes other than those shown in the electronic supplementary material, figure S4, in the paper were used to select data from for comparison to the experimental (laser speckle interferometry) data. While that model itself appears to be valid, the data taken from it are therefore invalid, as they do not match the location of the region that was experimentally sampled.

2011

In this study, we sought to construct the first reliable FE model of the stress of an adult Asian elephant (*Elephas maximus* Linnaeus 1758) in order to move towards a better understanding of the forms, functions, adaptability and mechanical properties of the trunk. Strength, loading and stiffness in three planes of load were measured, and the trunk was divided into three sections: strength, loading and stiffness in three planes of load (Table 1). We first test one model's precision by comparing a series of computed results with the data to the size and type of elements used in the geometric model. We then perform a sensitivity study on the model by determining the effects of varying its material properties. Finally, we compare the results of the experimental data obtained under prescribed loading conditions in the laboratory (using live speckle photography). We hope that this study will meet the methodological standards in medical applications of FEA by encouraging the inclusion of convergence test, sensitivity studies and validation analyses to the model validity.



2014

2018

24) Furthermore, FEA strain values have been found to converge when element numbers exceed 200 000 [23] to 800 000 [19] elements. Here, meshes comprised on average 2 million elements (mean 2 150 467, range 288 326–5 527 826). Meshes were exported from ABAQUS in the proprietary ABAQUS FEAP solver (Dassault Systems, Paris) file format.

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2026

complex interaction of their form-function relationships. But what makes a reliable subject-specific finite element model? To approach this question, we here present a set of convergence and sensitivity analyses and a validation study as an example, for finite element analysis (FEA) in general, of ways to create a reliable model. We detail four choices of element size, type and material properties in FEA to influence the results of stress analysis. We also detail the use of a heterogeneous material to help model material properties throughout an element (but of broad applicability to FEA). We then use an *in vivo* experimental validation test of a cadaveric femur to check our FEA results and find that the heterogeneous model matches the experimental results extremely well, and far better than the homogeneous model. This emphasizes how considering heterogeneous material properties in FEA may be critical, as this should become standard practice in comparative FEA studies along with convergence analysis. Consideration of element size and material properties in FEA may be required to obtain accurate results and derive reliable conclusions from them.

- In some papers, the citation is in the literature review
- In other paper, the paper findings are used to build upon

What if we *build our work* on involuntarily wrong or fraudulent results?
How can we *verify* that we and our collaborators produce correct results?
How can we *check the results* of the papers that we review?

We *need* to be able to:

Assess correctness of scientific claims

Build on reliable previous work

Compare and validate methods and algorithms

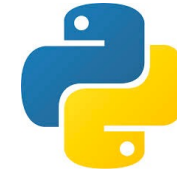
The *answers* to these *needs* are
Openness and Reproducibility

What are Openness and Reproducibility?

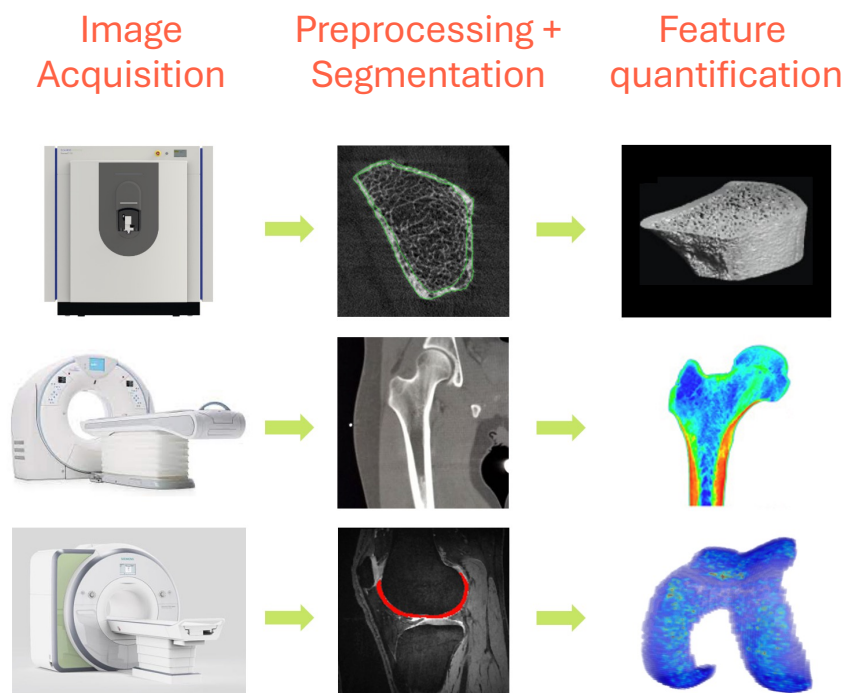
- **Openness** refers to the *free availability of data, software, and methods* developed by researchers with the aim to share knowledge and tools to professionals and citizens ([Woelfle 2011](#))
- **Reproducibility** is the ability of researchers to *duplicate the results of a previous study* using the same data, software, and methods used by the original authors ([Bollen 2015](#))

The open and reproducible research movement

- About 15-10 years ago, scientists started to question more and more the way they work
 - Bottom-up movement
- Creation/explosion of tools to make research open and reproducible
 - Zenodo, Jupyter notebook, ...
- Science is more and more intended as a collective effort
 - Creation of communities to solve specific needs



Needs in MSK image-based research



- Within labs, code is often **fragmented**
 - Code can be tied to the creator's **employment** → hard to reuse and extend
 - Proprietary software can limit understanding and customization
- Need for algorithm **comparisons** and **standardization** across labs
 - Faster advancement of research

The Open and Reproducible Musculoskeletal Imaging Research (**ORMIR**) Community

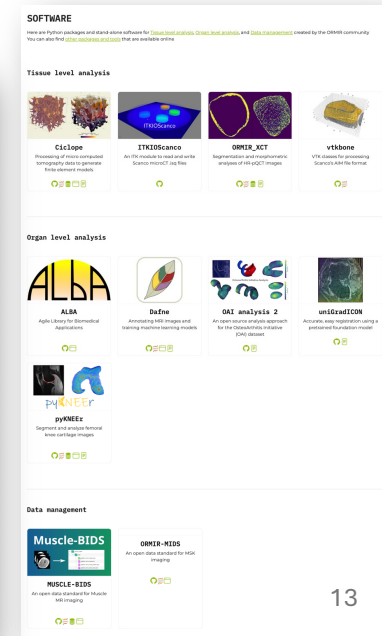
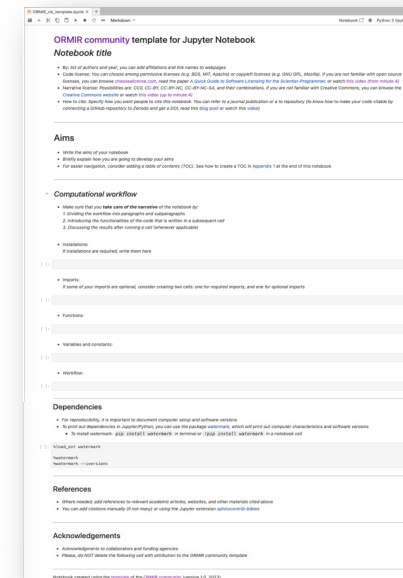
- More than 70 international researchers from industry and academia
- Goals
 - **Create** open, reproducible, well-tested, and well-documented **code** to analyze MSK images
 - **Standardize data acquisition and management** to favor data sharing and algorithm comparison
 - **Promote a culture of openness and reproducibility** in MSK imaging research for a faster advancement of the field
- Code of conduct
 - Inclusive, friendly, respectful, considerate



The Open and Reproducible Musculoskeletal Imaging Research (ORMIR) Community

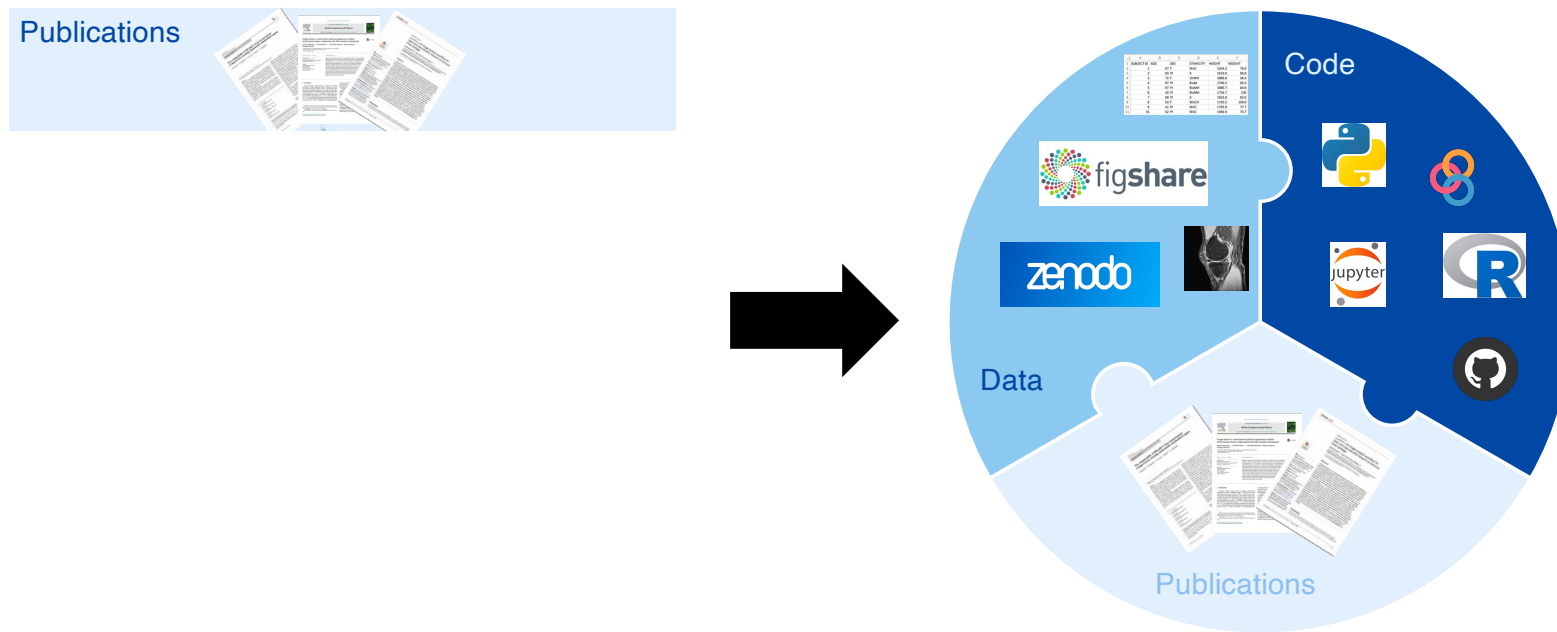


- 6 Working groups
 - Biomechanics, High-res CT, Knee, Muscle, ORMIR-MIDS, Spine
- Community and Technical Advisory Boards, Social Media Team
 - Guidelines and templates (ormir.org)
 - Follow us on LinkedIn
- Events
 - Workshops, events at conferences



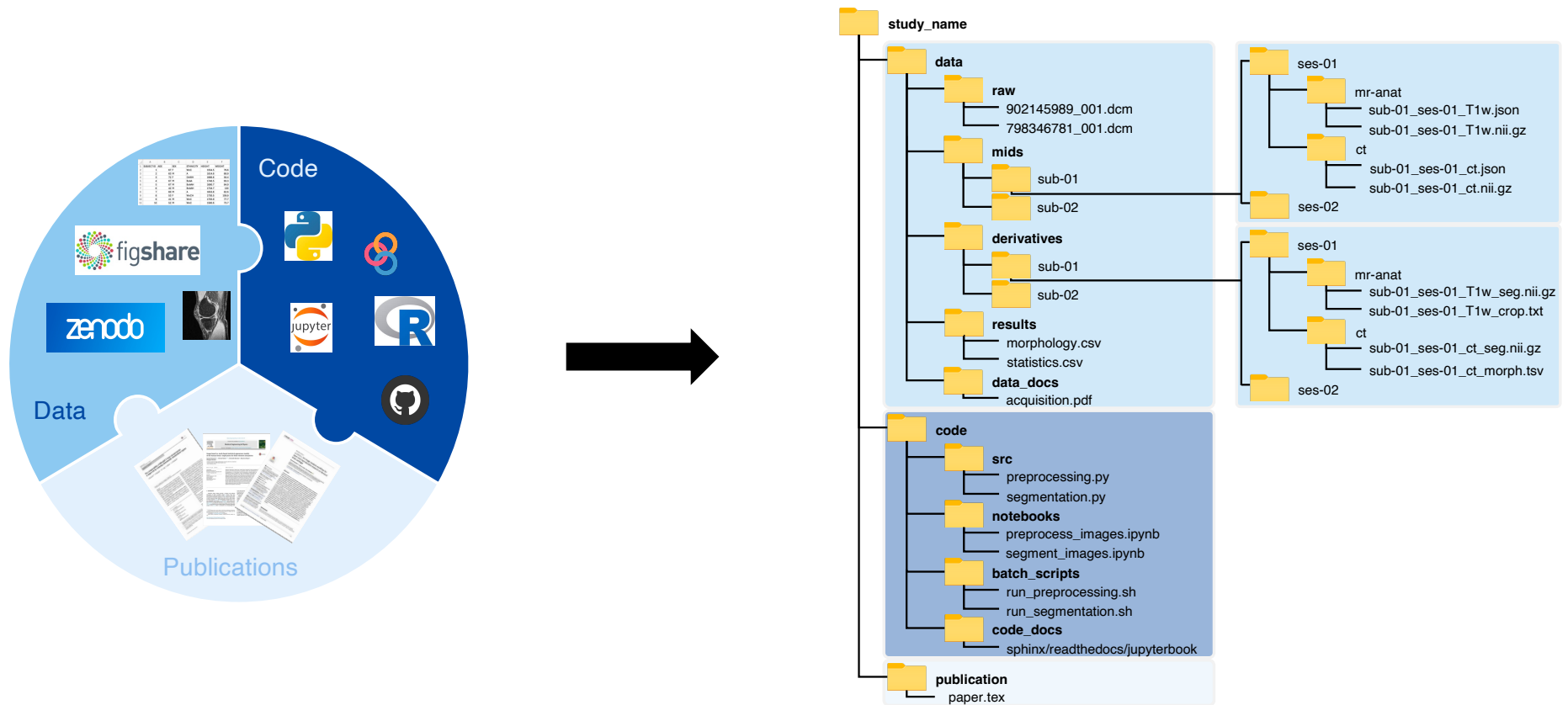
How can we do open and
reproducible research?

What's a research output?



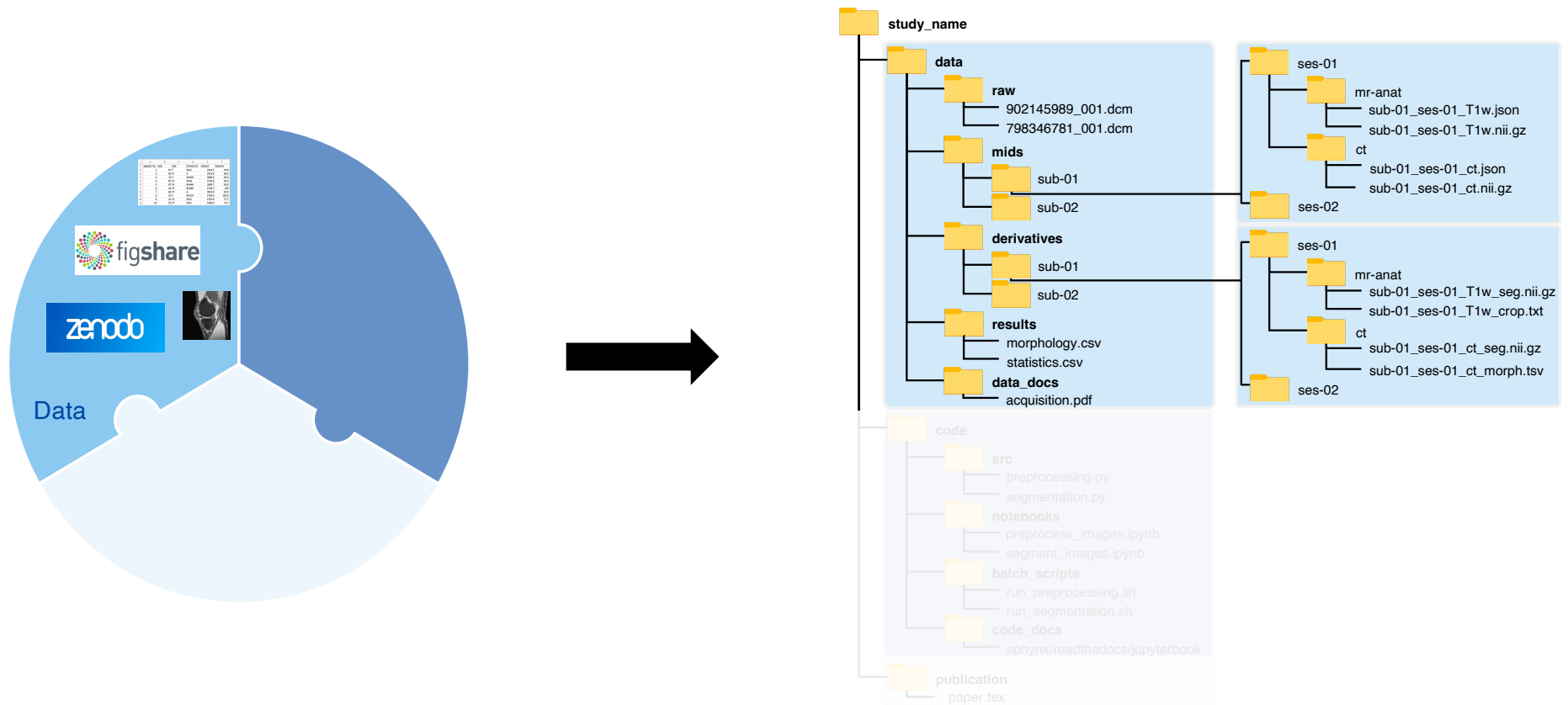
Open and reproducible research in musculoskeletal imaging: why it matters and how to implement it with the guidelines of the Open and Reproducible Musculoskeletal Imaging Research (ORMIR) community. JBMRPlus, 2026 (<https://doi.org/10.1093/jbmrpl/ziag025>)

How to **organize** a research project for **openness** and **reproducibility**?

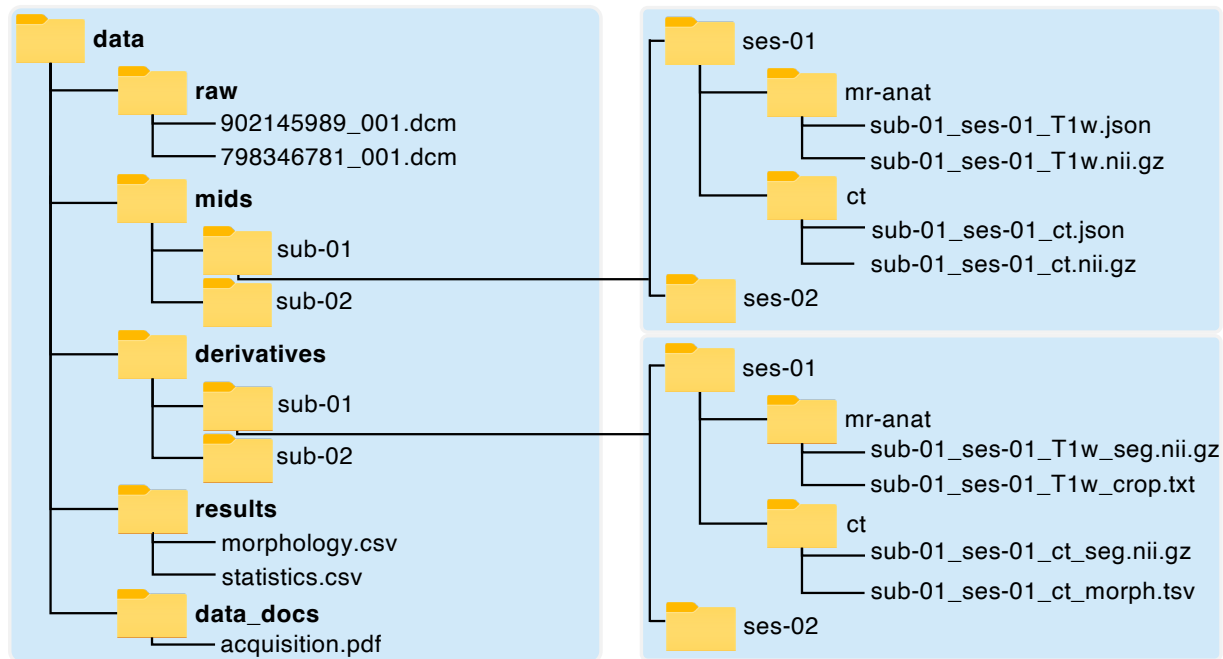
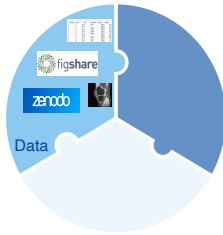


Find the template on www.ormir.org/templates.html!

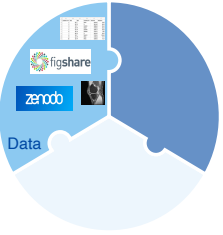
How to **organize** a research project for **openness** and **reproducibility**?



Data and metadata



- Data transformation
 - raw → mids using ORMIR-MIDS (<https://github.com/ormir-mids>)
 - mids → derivatives → results
- Tracking **provenance**
- Open file **formats**
 - E.g., .csv instead of .xlsx



Making the data open

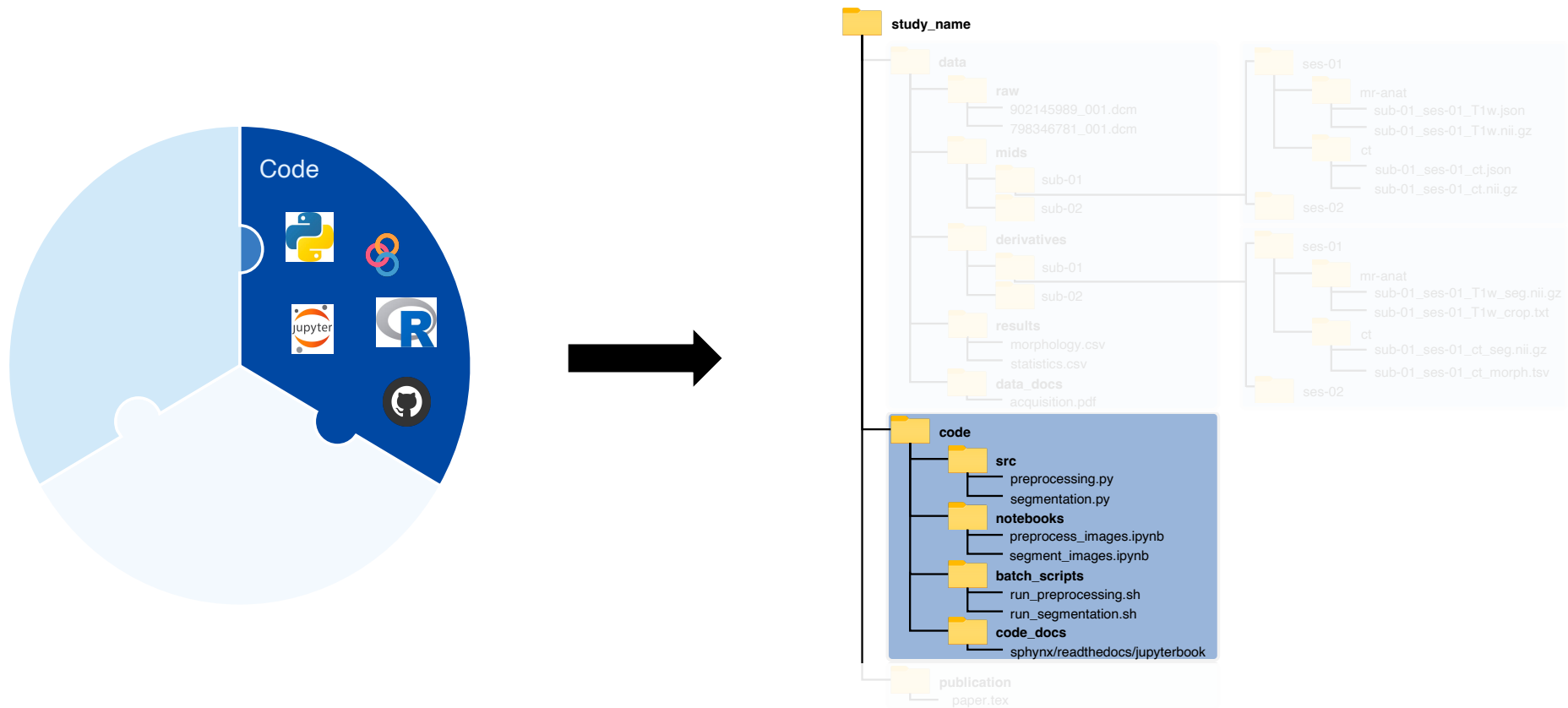
- Discuss sharing data with your [supervisors/institution!](#)
 - See the ORMIR Data Sharing Guidelines (www.ormir.org/data_sharing_guidelines/)
- At least 3 types of repositories:
 - [Institutional](#)
 - [Topic-specific](#) (E.g. openneuro.org)
 - [Generic](#) (E.g. Zenodo, Figshare)
- Why a data repository?
 - Link to *personal* repositories often get deleted
 - They provide a [DOI](#) → Citations
 - They often provide version control and embargo period
- [License](#)
 - Online material is automatically protected by copyright
 - Creative Common licenses / Institutional licenses

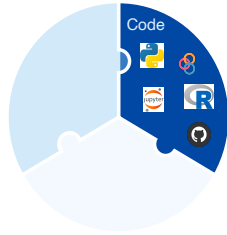
Version 3	Jan 14, 2020
10.5281/zenodo.3607230	
Version 2	Nov 13, 2019
10.5281/zenodo.3540477	
Version 1	Jun 29, 2019
10.5281/zenodo.3262308	
Cite all versions? You can cite all versions by using the DOI 10.5281/zenodo.3262307 . This DOI represents all versions, and will always resolve to the latest one. Read more.	



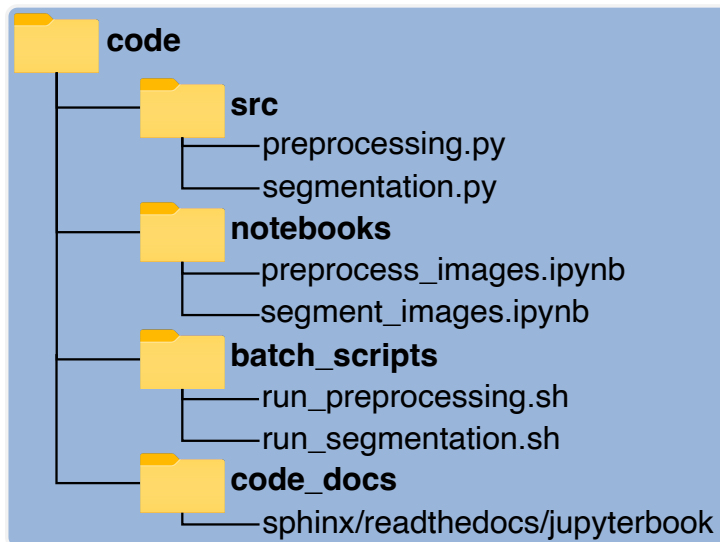
<https://creativecommons.org/about/ccllicenses/19>

How to **organize** a research project for **openness** and **reproducibility**?





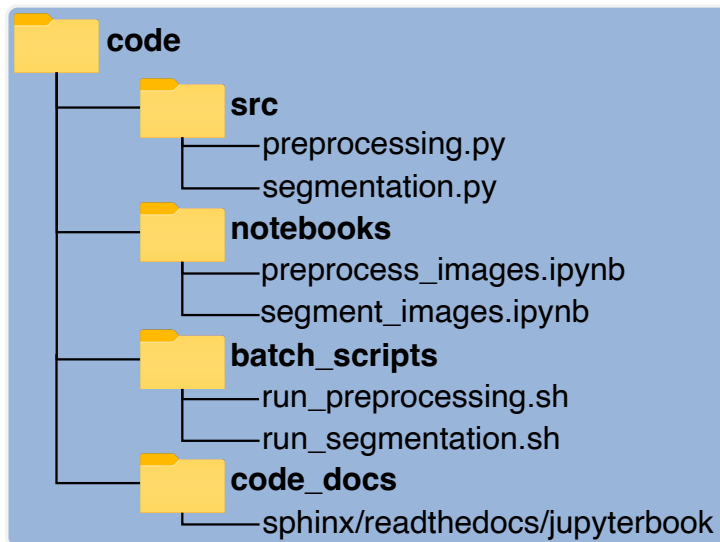
Code – Open-source



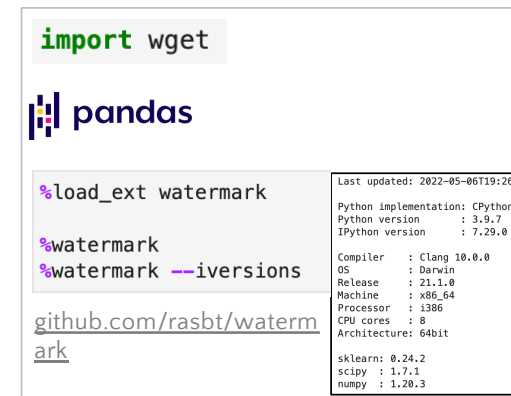
- Use [open-source](#) programming languages
 - [Python](#) (R, Julia, ...)
 - Look at the available packages in your field!



Code – Computational narratives



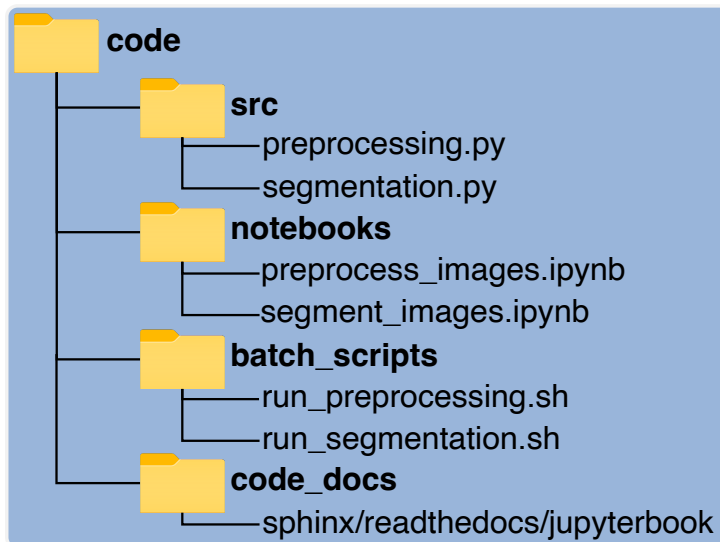
- **Reproducible workflows** in notebooks
 - Automatically download data from a repository
 - Automatic intervention for data analysis
 - Dependencies, i.e., versions of libraries and machine



Find the template on www.ormir.org/templates.html!



Code – Documentation



- Write **code** and **documentation** for **humans**!

- Style guide for Python Code

(<https://peps.python.org/pep-0008>)

- NumPy/Google guide for docstrings

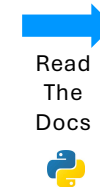
(<https://numpydoc.readthedocs.io/en/latest/format.html>)

```
def read_image(filename, format='SimpleITK'):
    """Reads an image from file as either a SimpleITK or ITK image. The default is SimpleITK

    Parameters
    -----
    filename : str
        Filename of the image to read (extension can be .nha, .mhd, etc.)
    format : str
        Format of the read image, defined by the user. It can be:
        - "SimpleITK": Reads the image using SimpleITK. It's the default
        - "ITK": Reads the image using ITK

    Returns
    -----
    img: SimpleITK or ITK
        Image in SimpleITK or ITK format
    """
```

docstring



```
def read_image(filename, format='SimpleITK')
    Reads an image from file as either a SimpleITK or ITK image. The default is SimpleITK

    Parameters
    -----
    filename : str
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        It's the default - "ITK": Reads the image using ITK

    Returns
    -----
    img : SimpleITK or ITK
        Image in SimpleITK or ITK format
```

HTML documentation



Version control and licenses

- Version control

- Tracking code versions and contributions
- GitHub/Zenodo allows code citations
- Find the template for the README.md file on www.ormir.org/templates.html



- Open source license

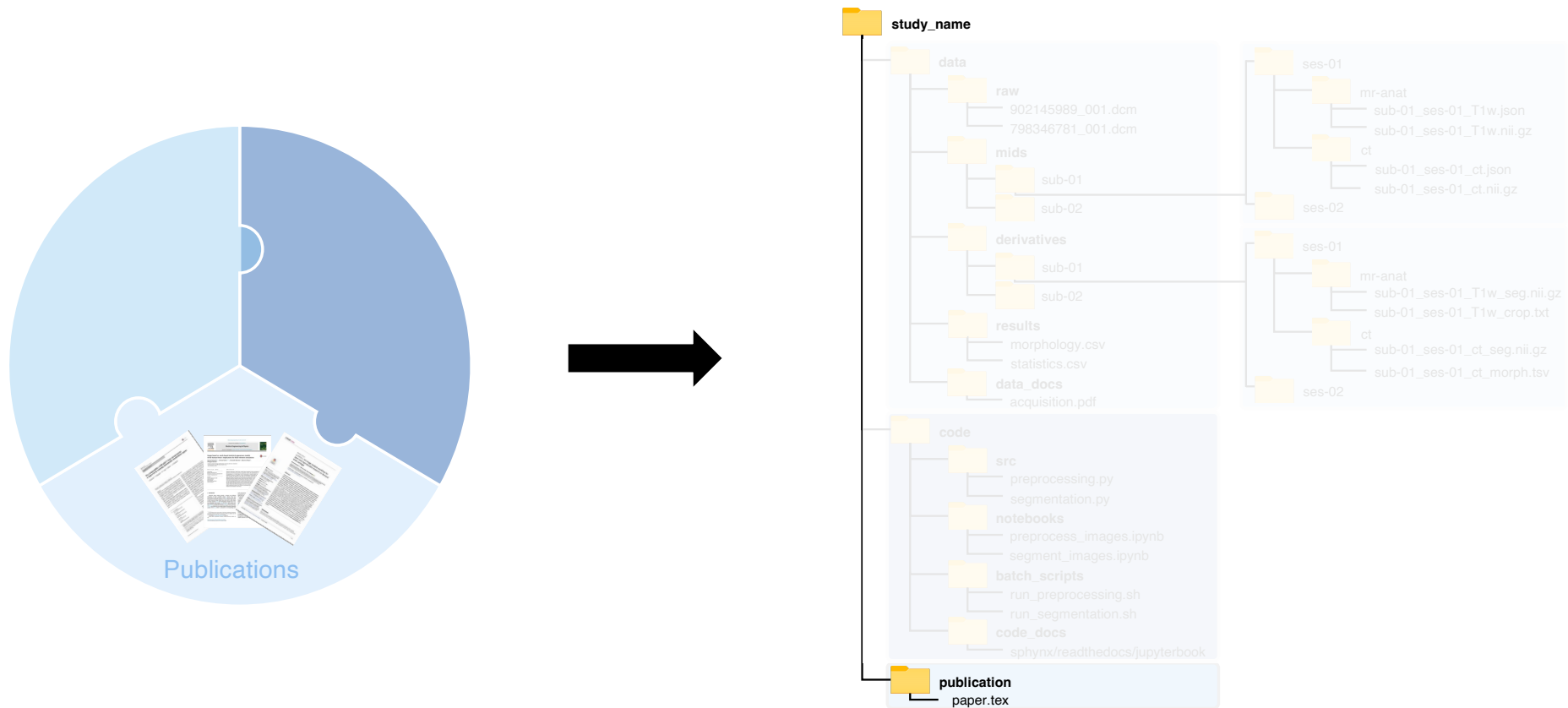
- Permissive / Restrictive



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BSD Zero Clause License	●	●	●		●						●		●
Academic Free License v3.0	●	●	●	●	●		●			●	●	●	●
GNU Affero General Public License v3.0	●	●	●	●	●	●	●	●	●	●	●		●
Apache License 2.0	●	●	●	●	●		●			●	●	●	●
Artistic License 2.0	●	●	●	●	●		●			●	●	●	●
BSD 2-Clause "Simplified" License	●	●	●		●		●				●		●
BSD 3-Clause Clear License	●	●	●	●	●		●				●		●
BSD 3-Clause "New" or "Revised" License	●	●	●		●		●				●		●
BSD 4-Clause "Original" or "Old" License	●	●	●		●		●				●		●
Boost Software License 1.0	●	●	●		●		●				●		●

<https://choosealicense.com/appendix/>

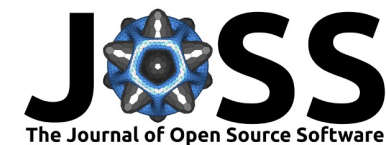
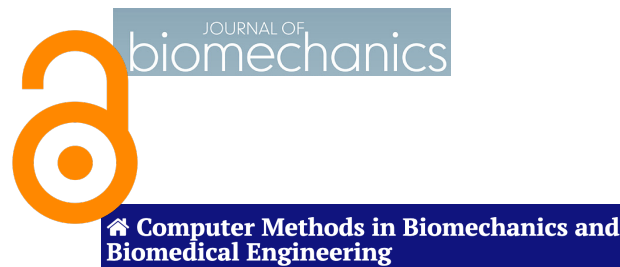
How to **organize** a research project for **openness** and **reproducibility**?





Sharing the reports

- Openly share our work reports
 - [Preprints](#) on the Archives or Zenodo/Figshare
 - [Open access](#) in traditional journals
 - [Specific journals](#) for code and data

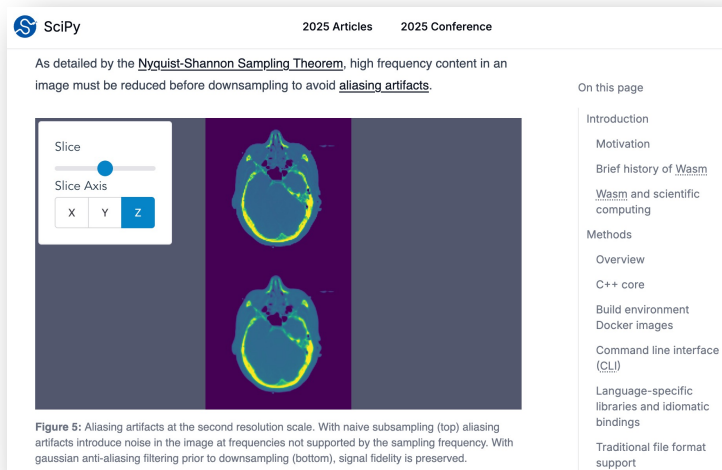




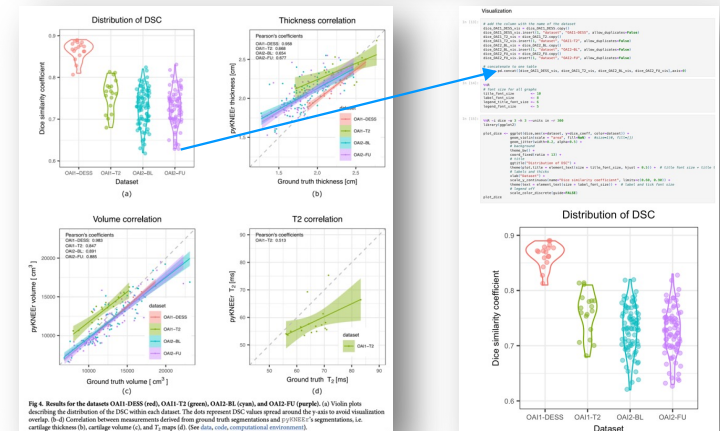
Towards reproducible publications

- Share information about **data** and **code**
- Reproducible **analysis** and **graphs**
- **SciPy proceedings** are Jupyter notebooks!
<https://proceedings.scipy.org/2025>

Software Used	Repository	Metadata / Documentation	Software / Language	License	DOI	Citation
Preprocessing	Bitbucket	Wiki	C++, ITK	Apache	https://doi.org/10.1016/j.media.2014.08.008	[59]*
elastix 4.8	GitHub	GitHub Wiki	C++, ITK	Apache	https://doi.org/10.5281/zenodo.2009.2035616	[28]*
Developed pyKNEER	GitHub	Website	python, Jupyter notebook	GNU GPLv3	https://doi.org/10.5281/zenodo.2574172	Bonaretti S. et al. "pyKNEER" (v0.0.1). Zenodo. 2019. 10.5281/zenodo.2574172
Data Original	OAI	Website	-	Data user agreement	https://doi.org/10.1016/j.joca.2008.06.016	[49]*
Derived (results)	Zenodo	Jupyter notebook	-	CC-BY-NC-SA	https://doi.org/10.5281/zenodo.2530609	Bonaretti S. et al. Dataset used in (Bonaretti et al. 2019). Zenodo. 2019. 10.5281/zenodo.2530609



McCormick 2024, <https://proceedings.scipy.org/articles/TCFJ5130>



Bonaretti 2020, <https://doi.org/10.1371/journal.pone.0226501>

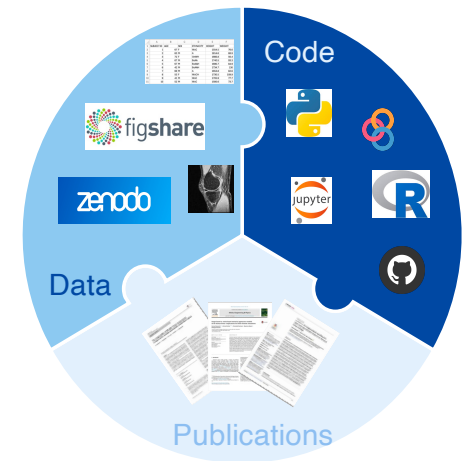
What now?

Some references

- Ten simple rules for writing and sharing computational analyses in Jupyter Notebooks (Rule et al, doi.org/10.1371/journal.pcbi.1007007)
- Ten simple rules for reproducible computational research (Sandve et al, doi.org/10.1371/journal.pcbi.1003285)
- Ten simple rules for taking advantage of Git and GitHub (Perez-Riverol et al, doi.org/10.1371/journal.pcbi.1004947)
- Good enough practices in scientific computing (Wilson et al, doi.org/10.1371/journal.pcbi.1005510)
- A quick guide to software licensing for the scientist-programmer (Morin et al, doi.org/10.1371/journal.pcbi.1002598)
- Toward the Geoscience paper of the future: Best practices for documenting and sharing research from data to software to provenance (Gil et al, doi.org/10.1002/2015EA000136)
- Jupyter: Thinking and storytelling with code and data (Granger & Peréz, doi.org/10.1109/MCSE.2021.3059263)

Take home message

- Open and reproducible research is a *way of working* that **supports researchers' needs!**
- Nowadays, **we have all the tools** to do open and reproducible research!
- Find plenty of **material** on www.ormir.org: templates, code and data guidelines, resources to learn more
- It is a **learning** process! One step at the time! Just...





Picture by Geralt (pixabay)